Listing 9.4 Using CloudFormation to launch an EC2 instance and mount an EFS filesystem

```
Resources:
  [...]
  EC2InstanceA:                                                    ①
    Type: 'AWS::EC2::Instance'
    Properties:
      ImageId: !FindInMap [RegionMap, !Ref 'AWS::Region', AMI]
      InstanceType: 't2.micro'
      IamInstanceProfile: !Ref IamInstanceProfile                  ②
      NetworkInterfaces:
      - AssociatePublicIpAddress: true
        DeleteOnTermination: true
        DeviceIndex: 0
        GroupSet:
        - !Ref EFSClientSecurityGroup                              ③
        SubnetId: !Ref SubnetA                                     ④
      UserData:                                                    ⑤
        'Fn::Base64': !Sub |
          #!/bin/bash -ex
          trap '/opt/aws/bin/cfn-signal -e 1 --stack ${AWS::StackName}
➥ --resource EC2InstanceA --region ${AWS::Region}' ERR

          # install dependencies
          yum install -y nc amazon-efs-utils
          pip3 install botocore

          # copy existing /home to /oldhome
          mkdir /oldhome
          cp -a /home/. /oldhome                                   ⑥

          # wait for EFS mount target
          while ! (echo > /dev/tcp/${FileSystem}.efs.${AWS::Region}.
➥ amazonaws.com/2049) >/dev/null 2>&1; do sleep 5; done           ⑦

          # mount EFS filesystem
          echo "${FileSystem}:/ /home efs _netdev,noresvport,tls,iam 0 0"
➥ >> /etc/fstab                                                   ⑧
          mount -a                                                 ⑨

          # copy /oldhome to new /home
          cp -a /oldhome/. /home                                   ⑩

          /opt/aws/bin/cfn-signal -e $? --stack ${AWS::StackName}
➥ --resource EC2InstanceA --region ${AWS::Region}
      Tags:
```